

Top-3 ensemble analysis and research reporting

P16 Exhaustive three-model enumeration

SYSTEM	This selection stage is deterministic. It must reuse stored predictions, the three dataset evaluation bundles and accepted 50-round weights without changing the metric protocol or adding model-specific bonuses.
USER	Enumerate every unique three-model set, align samples, calculate equal-probability soft voting and score cross-dataset performance and stability using the stored evidence.
VERIFIED OUTPUT	816 unique three-model combinations evaluated
SOURCE	ensemble_top3_selector.py · data/results_manual/ensemble_top3_combination_ranking.csv

P17 Complementarity recommendation

SYSTEM	Prefer combinations supported by performance, stability and complementary errors. Do not claim leakage-free superiority while dataset independence or homology gates remain unresolved.
USER	Select one auditable trio from the complete combination ranking and persist the method, datasets, candidate models, metrics, ranking evidence and scientific caveat.
VERIFIED OUTPUT	Recommended trio: C_AMPs-predict / esm-AxP-GDL / pepnet_standard
SOURCE	data/results_manual/ensemble_top3_selection.json

P18 Research Advisor synthesis

SYSTEM	Write a concise report grounded only in final weights, ranking and dataset profiles. Clearly distinguish exploratory ranking from leakage-free formal validation. Explain Top-3, weight consensus, score IQR and Top-3 frequency.
USER	Analyze cross-dataset generalization, model limitations, dynamic metrics, next-generation AMP model directions, benchmark construction and experiment priorities. Missing evidence must be stated explicitly.
VERIFIED OUTPUT	amp_future_directions_report_codex_agents.md generated from stored benchmark evidence
SOURCE	agents/weight_meeting/research_advisor.md · agents/runtime_prompts/amp_research_advisor_template.md

P19 Publication and audit handoff

SYSTEM	Figures are a reporting layer and cannot feed values back into ranking. Every conclusion must remain traceable to JSON, CSV, Markdown, source code and run manifests.
USER	Export editable vector figures, high-resolution previews, source tables, figure contracts and QA notes. Keep exploratory conclusions and external-validation requirements visible in captions and reports.
VERIFIED OUTPUT	SVG/PDF/PNG figures, source CSVs, reports and complete Agent audit records preserved
SOURCE	figures/ · data/results_manual/ · run_manifest.py · tests/